



EM Algorithm

Visualization

On the webpage <http://jormungand.net/projects/misc/em/> you can find “cluster demo (javascript version)”.

“Paint” different data sets and run the EM and k-means algorithms for different parameters k to get a feeling, how they work.

- Press “clr”
- Paint some data
- Pick the parameter k by pressing one of the numbers 1 to 9
- Click either km (for k-means) or em (for EM) MULTIPLE times, to see how the centroids move

Note that EM only shows the “hard” assignments of the data points. Actually, each point is assigned to every cluster!

- Try gaussian and non-gaussian clusters.
- Try k -values that are higher than the number of clusters you painted.
- Try k -values that are lower than the number of clusters you painted.

Hint: May not work in every Browser (Chrome is fine)